NYPD Shooting Incident Data Report

KC

2026-06-05

```
# Load necessary packages:
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.3

## Warning: package 'readr' was built under R version 4.5.3

## Warning: package 'forcats' was built under R version 4.5.3

## Warning: package 'lubridate' was built under R version 4.5.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
```

Project Step 1: Start Rmd Document

This project will use a dataset of shooting incidents in New York City. I wanted to look at if victim age group and borough are associated with differences in shooting incident counts around New York City.

We will need to download the NYPD Crime dataset called 'ARCHIVED_NYPD Shooting Incident Data (Historic)' on data.gov

Getting current data

Download csv from a url and save the csv data to a variable.

```
url_in <- "https://data.cityofnewyork.us/api/views/833y-fsy8/rows.csv?accessType=DOWNLOAD"

mydata <- read_csv(url_in)
```

```
## Rows: 29744 Columns: 21
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (12): OCCUR_DATE, BORO, LOC_OF_OCCUR_DESC, LOC_CLASSFCTN_DESC, LOCATION...
## dbl   (5): INCIDENT_KEY, PRECINCT, JURISDICTION_CODE, Latitude, Longitude
## num   (2): X_COORD_CD, Y_COORD_CD
## lgl   (1): STATISTICAL_MURDER_FLAG
## time  (1): OCCUR_TIME
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
mydata
```

```
## # A tibble: 29,744 x 21
##    INCIDENT_KEY OCCUR_DATE OCCUR_TIME BORO     LOC_OF_OCCUR_DESC PRECINCT
##           <dbl> <chr>      <time>     <chr>    <chr>                <dbl>
##  1    231974218 08/09/2021 01:06      BRONX    <NA>                    40
##  2    177934247 04/07/2018 19:48      BROOKLYN <NA>                    79
##  3    255028563 12/02/2022 22:57      BRONX    OUTSIDE                 47
##  4     25384540 11/19/2006 01:50      BROOKLYN <NA>                    66
##  5     72616285 05/09/2010 01:58      BRONX    <NA>                    46
##  6     85875439 07/22/2012 21:35      BRONX    <NA>                    42
##  7     79780323 07/12/2011 22:26      BROOKLYN <NA>                    71
##  8     85744504 07/14/2012 23:45      BROOKLYN <NA>                    69
##  9    142324890 04/21/2015 15:36      BROOKLYN <NA>                    75
## 10    152868707 05/07/2016 15:23      BROOKLYN <NA>                    69
## # i 29,734 more rows
## # i 15 more variables: JURISDICTION_CODE <dbl>, LOC_CLASSFCTN_DESC <chr>,
## #   LOCATION_DESC <chr>, STATISTICAL_MURDER_FLAG <lgl>, PERP_AGE_GROUP <chr>,
## #   PERP_SEX <chr>, PERP_RACE <chr>, VIC_AGE_GROUP <chr>, VIC_SEX <chr>,
## #   VIC_RACE <chr>, X_COORD_CD <dbl>, Y_COORD_CD <dbl>, Latitude <dbl>,
## #   Longitude <dbl>, Lon_Lat <chr>
```

Project Step 2: Tidy and Transform Data

Initial summary of the data

```
summary(mydata)
```

```
##   INCIDENT_KEY        OCCUR_DATE          OCCUR_TIME
##  Min.   :  9953245   Length:29744       Min.   :00:00:00.000000
##  1st Qu.: 67321140   Class :character   1st Qu.:03:30:45.000000
##  Median :109291972   Mode  :character   Median :15:15:00.000000
##  Mean   :133850951                      Mean   :12:46:10.874798
##  3rd Qu.:214741917                      3rd Qu.:20:44:00.000000
##  Max.   :299462478                      Max.   :23:59:00.000000
##
##      BORO           LOC_OF_OCCUR_DESC     PRECINCT      JURISDICTION_CODE
##  Length:29744       Length:29744       Min.   :  1.00   Min.   :0.0000
##  Class :character   Class :character   1st Qu.: 44.00   1st Qu.:0.0000
```

```
##  Mode  :character   Mode  :character   Median : 67.00   Median :0.0000
##                                        Mean   : 65.23   Mean   :0.3181
##                                        3rd Qu.: 81.00   3rd Qu.:0.0000
##                                        Max.   :123.00   Max.   :2.0000
##                                                          NA's   :2
##  LOC_CLASSFCTN_DESC LOCATION_DESC      STATISTICAL_MURDER_FLAG
##  Length:29744       Length:29744       Mode :logical
##  Class :character   Class :character   FALSE:23979
##  Mode  :character   Mode  :character   TRUE :5765
##
##
##
##
##  PERP_AGE_GROUP       PERP_SEX          PERP_RACE         VIC_AGE_GROUP
##  Length:29744       Length:29744       Length:29744       Length:29744
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##    VIC_SEX            VIC_RACE           X_COORD_CD        Y_COORD_CD
##  Length:29744       Length:29744       Min.   : 914928   Min.   :125757
##  Class :character   Class :character   1st Qu.:1000094   1st Qu.:183042
##  Mode  :character   Mode  :character   Median :1007826   Median :195506
##                                        Mean   :1009442   Mean   :208722
##                                        3rd Qu.:1016739   3rd Qu.:239980
##                                        Max.   :1066815   Max.   :271128
##
##     Latitude       Longitude        Lon_Lat
##  Min.   :40.51   Min.   :-74.25   Length:29744
##  1st Qu.:40.67   1st Qu.:-73.94   Class :character
##  Median :40.70   Median :-73.91   Mode  :character
##  Mean   :40.74   Mean   :-73.91
##  3rd Qu.:40.83   3rd Qu.:-73.88
##  Max.   :40.91   Max.   :-73.70
##  NA's   :97      NA's   :97
```

Change appropriate variables to factor and date types and remove un-needed columns

You can see column OCCUR_DATE is of 'chr' type and needs to be converted to a date (we can use lubridate package).

```r
str(mydata)
```

```
## spc_tbl_ [29,744 x 21] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ INCIDENT_KEY           : num [1:29744] 2.32e+08 1.78e+08 2.55e+08 2.54e+07 7.26e+07 ...
##  $ OCCUR_DATE             : chr [1:29744] "08/09/2021" "04/07/2018" "12/02/2022" "11/19/2006" ...
##  $ OCCUR_TIME             : 'hms' num [1:29744] 01:06:00 19:48:00 22:57:00 01:50:00 ...
##   ..- attr(*, "units")= chr "secs"
##  $ BORO                   : chr [1:29744] "BRONX" "BROOKLYN" "BRONX" "BROOKLYN" ...
##  $ LOC_OF_OCCUR_DESC      : chr [1:29744] NA NA "OUTSIDE" NA ...
##  $ PRECINCT               : num [1:29744] 40 79 47 66 46 42 71 69 75 69 ...
```

```
##  $ JURISDICTION_CODE      : num [1:29744] 0 0 0 0 0 2 0 2 0 0 ...
##  $ LOC_CLASSFCTN_DESC     : chr [1:29744] NA NA "STREET" NA ...
##  $ LOCATION_DESC          : chr [1:29744] NA NA "GROCERY/BODEGA" "PVT HOUSE" ...
##  $ STATISTICAL_MURDER_FLAG: logi [1:29744] FALSE TRUE FALSE TRUE TRUE FALSE ...
##  $ PERP_AGE_GROUP         : chr [1:29744] NA "25-44" "(null)" "UNKNOWN" ...
##  $ PERP_SEX               : chr [1:29744] NA "M" "(null)" "U" ...
##  $ PERP_RACE              : chr [1:29744] NA "WHITE HISPANIC" "(null)" "UNKNOWN" ...
##  $ VIC_AGE_GROUP          : chr [1:29744] "18-24" "25-44" "25-44" "18-24" ...
##  $ VIC_SEX                : chr [1:29744] "M" "M" "M" "M" ...
##  $ VIC_RACE               : chr [1:29744] "BLACK" "BLACK" "BLACK" "BLACK" ...
##  $ X_COORD_CD             : num [1:29744] 1006343 1000083 1020691 985107 1009854 ...
##  $ Y_COORD_CD             : num [1:29744] 234270 189065 257125 173350 247503 ...
##  $ Latitude               : num [1:29744] 40.8 40.7 40.9 40.6 40.8 ...
##  $ Longitude              : num [1:29744] -73.9 -73.9 -73.9 -74 -73.9 ...
##  $ Lon_Lat                : chr [1:29744] "POINT (-73.92019278899994 40.80967347200004)" "POINT (-73
##  - attr(*, "spec")=
##   .. cols(
##   ..   INCIDENT_KEY = col_double(),
##   ..   OCCUR_DATE = col_character(),
##   ..   OCCUR_TIME = col_time(format = ""),
##   ..   BORO = col_character(),
##   ..   LOC_OF_OCCUR_DESC = col_character(),
##   ..   PRECINCT = col_double(),
##   ..   JURISDICTION_CODE = col_double(),
##   ..   LOC_CLASSFCTN_DESC = col_character(),
##   ..   LOCATION_DESC = col_character(),
##   ..   STATISTICAL_MURDER_FLAG = col_logical(),
##   ..   PERP_AGE_GROUP = col_character(),
##   ..   PERP_SEX = col_character(),
##   ..   PERP_RACE = col_character(),
##   ..   VIC_AGE_GROUP = col_character(),
##   ..   VIC_SEX = col_character(),
##   ..   VIC_RACE = col_character(),
##   ..   X_COORD_CD = col_number(),
##   ..   Y_COORD_CD = col_number(),
##   ..   Latitude = col_double(),
##   ..   Longitude = col_double(),
##   ..   Lon_Lat = col_character()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```
mydata$OCCUR_DATE <- mdy(mydata$OCCUR_DATE)
```

I am not interested in exact coordinates or latitude/longitude of the crime, so I will remove them.

```
mydata <- mydata[, !(names(mydata) %in% c(
  "X_COORD_CD",
  "Y_COORD_CD",
  "Latitude",
  "Longitude",
  "Lon_Lat"
))]
```

All columns need to be changed to factors except for: INCIDENT_KEY (no need to change since it is an ID), OCCUR_TIME (already a time datatype), STATISTICAL_MURDER_FLAG (okay due to is true/false).

```
mydata$BORO <- factor(mydata$BORO)
mydata$LOC_OF_OCCUR_DESC <- factor(mydata$LOC_OF_OCCUR_DESC)
mydata$PRECINCT <- factor(mydata$PRECINCT)
mydata$JURISDICTION_CODE <- factor(mydata$JURISDICTION_CODE)
mydata$LOC_CLASSFCTN_DESC <- factor(mydata$LOC_CLASSFCTN_DESC)
mydata$LOCATION_DESC <- factor(mydata$LOCATION_DESC)
mydata$PERP_AGE_GROUP <- factor(mydata$PERP_AGE_GROUP)
mydata$PERP_SEX <- factor(mydata$PERP_SEX)
mydata$PERP_RACE <- factor(mydata$PERP_RACE)
mydata$VIC_AGE_GROUP <- factor(mydata$VIC_AGE_GROUP)
mydata$VIC_SEX <- factor(mydata$VIC_SEX)
mydata$VIC_RACE <- factor(mydata$VIC_RACE)
```

Handle missing data

Let us re-check our datatypes and see a summary of the data now that we've done some initial tidying.

From the summary output, it looks like we have missing values in LOC_CLASSFCTN_DESC, LOCATION_DESC, PERP_AGE_GROUP, PERP_SEX, PERP_RACE.

My plan to handle missing values is as follows: The 3 'PERP' columns all have 1628 nulls, which the large number could make sense since we wouldn't know the details of a perpetrator that has not been caught. That being said, these columns have existing values for 'UNKNOWN' so I would likely replace these nulls with 'UNKNOWN'. The LOC_CLASSFCTN_DESC has over 2000 nulls but this column also has 'NA' values. I could replace the nulls with NA, assuming the NA means 'No answer' (I would have to confirm this before I actually replace the nulls, however.)

```
str(mydata)
```

```
## tibble [29,744 x 16] (S3: tbl_df/tbl/data.frame)
##  $ INCIDENT_KEY           : num [1:29744] 2.32e+08 1.78e+08 2.55e+08 2.54e+07 7.26e+07 ...
##  $ OCCUR_DATE             : Date[1:29744], format: "2021-08-09" "2018-04-07" ...
##  $ OCCUR_TIME             : 'hms' num [1:29744] 01:06:00 19:48:00 22:57:00 01:50:00 ...
##   ..- attr(*, "units")= chr "secs"
##  $ BORO                   : Factor w/ 5 levels "BRONX","BROOKLYN",..: 1 2 1 2 1 1 2 2 2 2 ...
##  $ LOC_OF_OCCUR_DESC      : Factor w/ 2 levels "INSIDE","OUTSIDE": NA NA 2 NA NA NA NA NA NA NA ...
##  $ PRECINCT               : Factor w/ 77 levels "1","5","6","7",..: 23 51 30 39 29 25 44 42 47 42 ..
##  $ JURISDICTION_CODE      : Factor w/ 3 levels "0","1","2": 1 1 1 1 1 3 1 3 1 1 ...
##  $ LOC_CLASSFCTN_DESC     : Factor w/ 10 levels "(null)","COMMERCIAL",..: NA NA 8 NA NA NA NA NA NA N
##  $ LOCATION_DESC          : Factor w/ 40 levels "(null)","ATM",..: NA NA 18 29 25 26 NA 26 25 NA ...
##  $ STATISTICAL_MURDER_FLAG: logi [1:29744] FALSE TRUE FALSE TRUE TRUE FALSE ...
##  $ PERP_AGE_GROUP         : Factor w/ 12 levels "(null)","<18",..: NA 8 1 12 8 5 NA NA 8 5 ...
##  $ PERP_SEX               : Factor w/ 4 levels "(null)","F","M",..: NA 3 1 4 3 3 NA NA 3 3 ...
##  $ PERP_RACE              : Factor w/ 8 levels "(null)","AMERICAN INDIAN/ALASKAN NATIVE",..: NA 8 1 6
##  $ VIC_AGE_GROUP          : Factor w/ 7 levels "<18","1022","18-24",..: 3 4 4 3 1 3 4 4 4 3 ...
##  $ VIC_SEX                : Factor w/ 3 levels "F","M","U": 2 2 2 2 1 2 2 2 2 2 ...
##  $ VIC_RACE               : Factor w/ 7 levels "AMERICAN INDIAN/ALASKAN NATIVE",..: 3 3 3 3 3 3 3 7 3
```

```
summary(mydata)
```

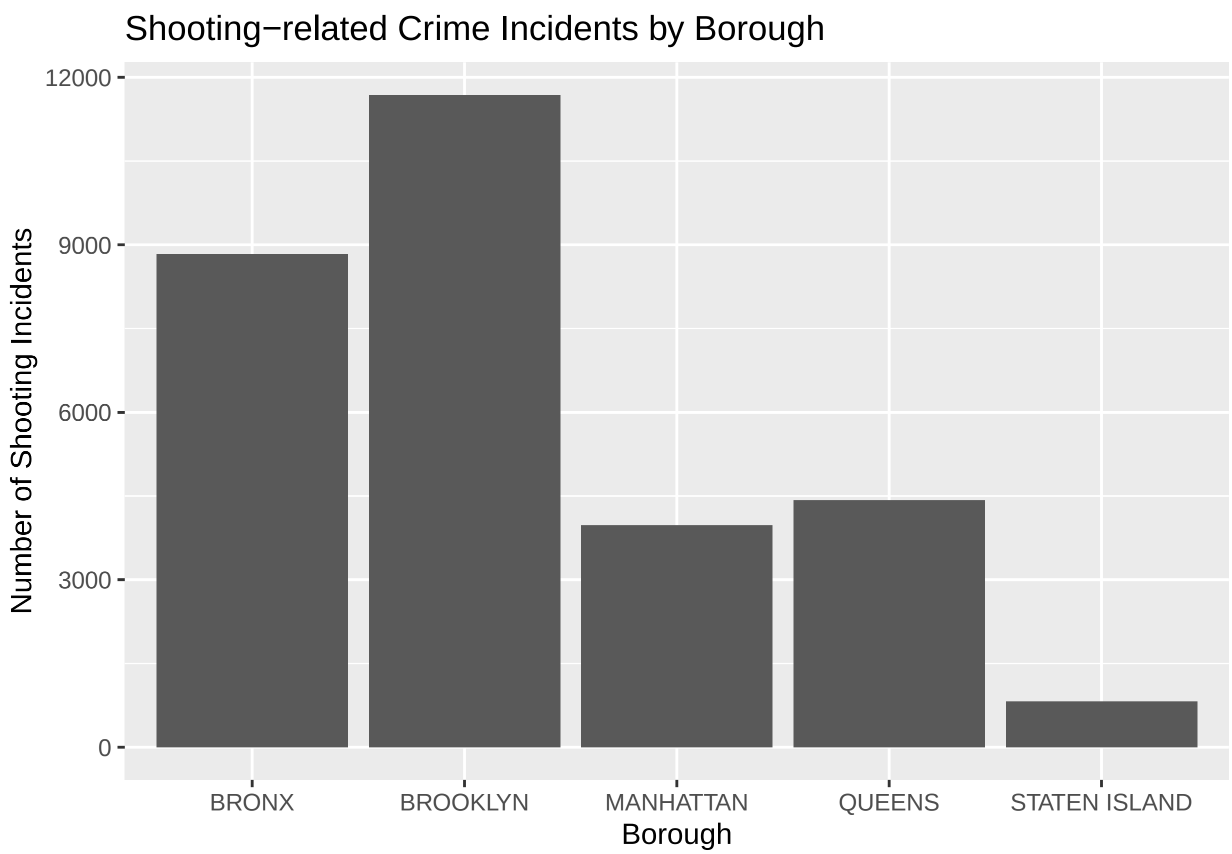
```
##   INCIDENT_KEY          OCCUR_DATE            OCCUR_TIME
```

```
##  Min.   :  9953245   Min.   :2006-01-01   Min.   :00:00:00.000000
##  1st Qu.: 67321140   1st Qu.:2009-10-29   1st Qu.:03:30:45.000000
##  Median :109291972   Median :2014-03-25   Median :15:15:00.000000
##  Mean   :133850951   Mean   :2014-10-31   Mean   :12:46:10.874798
##  3rd Qu.:214741917   3rd Qu.:2020-06-29   3rd Qu.:20:44:00.000000
##  Max.   :299462478   Max.   :2024-12-31   Max.   :23:59:00.000000
##
##            BORO       LOC_OF_OCCUR_DESC    PRECINCT     JURISDICTION_CODE
##  BRONX         : 8834   INSIDE :  682     75     : 1680   0    :24957
##  BROOKLYN      :11685   OUTSIDE: 3466     73     : 1561   1    :   109
##  MANHATTAN     : 3977   NA's   :25596     67     : 1288   2    : 4676
##  QUEENS        : 4426                     44     : 1159   NA's:    2
##  STATEN ISLAND:  822                      79     : 1073
##                                           47     : 1048
##                                           (Other):21935
##   LOC_CLASSFCTN_DESC                   LOCATION_DESC   STATISTICAL_MURDER_FLAG
##  STREET     : 2639    MULTI DWELL - PUBLIC HOUS: 5188   Mode :logical
##  HOUSING    :  643    MULTI DWELL - APT BUILD  : 3042   FALSE:23979
##  DWELLING   :  341    (null)                   : 2526   TRUE :5765
##  COMMERCIAL:  276    PVT HOUSE                : 1010
##  OTHER      :   74    GROCERY/BODEGA           :  775
##  (Other)    :  175    (Other)                  : 2226
##  NA's       :25596    NA's                     :14977
##  PERP_AGE_GROUP   PERP_SEX            PERP_RACE     VIC_AGE_GROUP   VIC_SEX
##  18-24  :6630   (null): 1628   BLACK         :12323   <18     : 3081   F: 2891
##  25-44  :6342   F     :  461   WHITE HISPANIC: 2667   1022    :     1   M:26841
##  UNKNOWN:3148   M     :16845   UNKNOWN       : 1838   18-24  :10677   U:   12
##  <18    :1805   U     : 1500   (null)        : 1628   25-44  :13563
##  (null) :1628   NA's  : 9310   BLACK HISPANIC: 1487   45-64  : 2118
##  (Other): 847                  (Other)       :  491   65+    :  236
##  NA's   :9344                  NA's          : 9310   UNKNOWN:   68
##                            VIC_RACE
##  AMERICAN INDIAN/ALASKAN NATIVE:   13
##  ASIAN / PACIFIC ISLANDER      :  478
##  BLACK                         :20999
##  BLACK HISPANIC                : 2930
##  UNKNOWN                       :   72
##  WHITE                         :  741
##  WHITE HISPANIC                : 4511
```

Project Step 3: Add Visualizations & Analysis

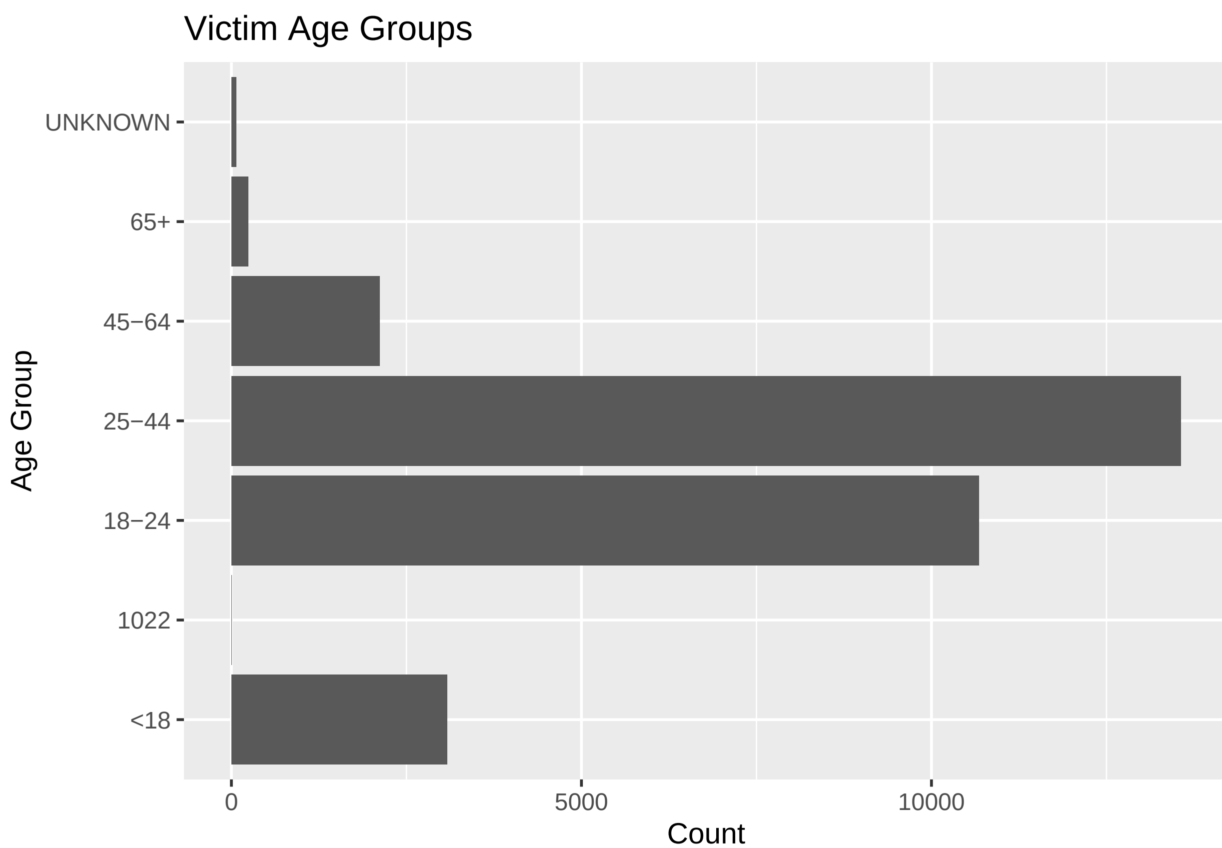
This bar chart shows how much shooting-related crime is happening in each borough. Visually we can see Brooklyn has the most crime incidents and Staten Island the least.

```r
mydata %>%
  ggplot(aes(x = BORO)) +
  geom_bar() +
  labs(title = "Shooting-related Crime Incidents by Borough",
       x = "Borough",
       y = "Number of Shooting Incidents")
```



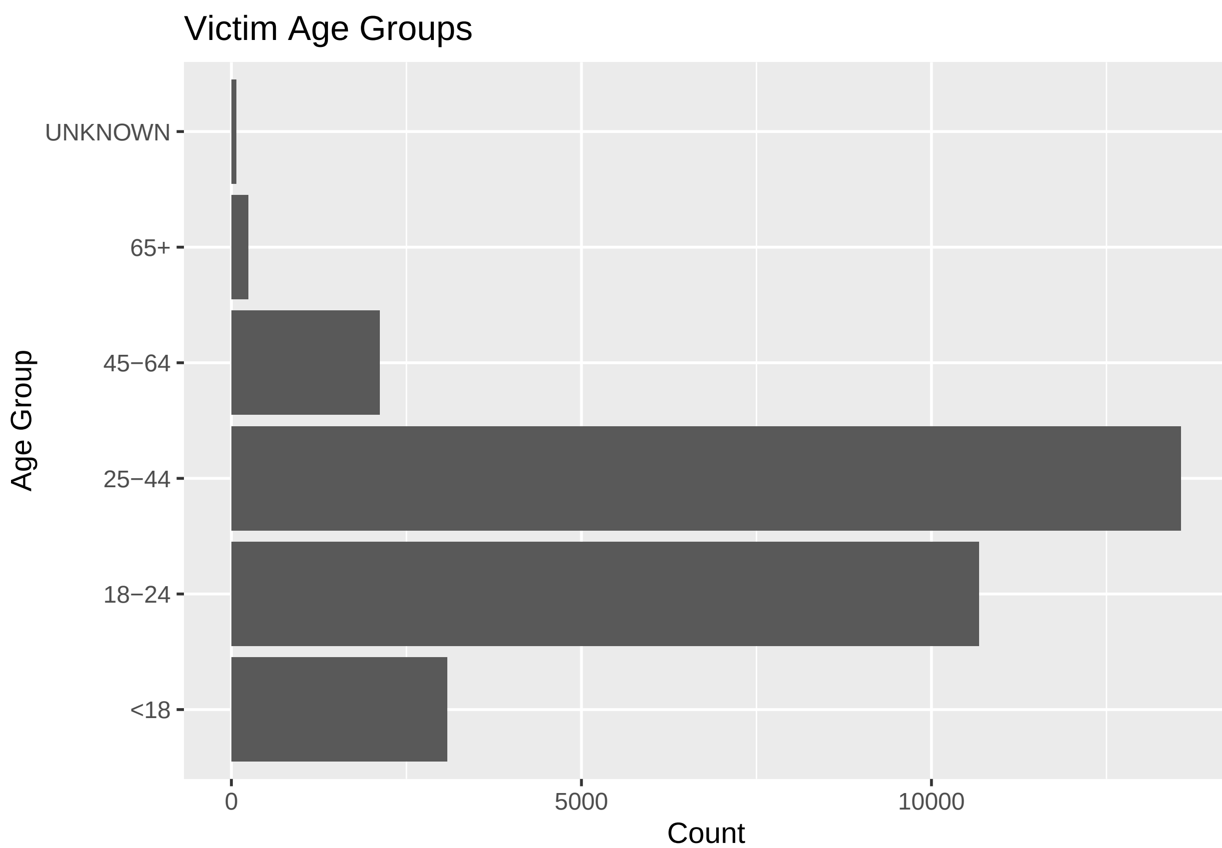
I also wanted to see what age was most likely to be a victim. From the chart below, we can see the most victimized group was the ages between 25-44.

```
mydata %>%
  ggplot(aes(x = VIC_AGE_GROUP)) +
  geom_bar() +
  coord_flip() +
  labs(title = "Victim Age Groups",
       x = "Age Group",
       y = "Count")
```



There is also an age group called '1022'; looking back at the summary(mydata) there is only one record with this value so it must be an error. I will filter this data to show only valid age group categories in an updated visualiation below.

```
mydata_clean <- mydata %>%
  filter(VIC_AGE_GROUP != "1022")

mydata_clean %>%
  ggplot(aes(x = VIC_AGE_GROUP)) +
  geom_bar() +
  coord_flip() +
  labs(title = "Victim Age Groups",
       x = "Age Group",
       y = "Count")
```



Some more exploration and analysis

Let us see how age is distributed in the boroughs.

```
age_boro <- mydata_clean %>%
  group_by(BORO, VIC_AGE_GROUP) %>%
  summarize(count = n(), .groups = "drop")

print(age_boro, n = Inf)
```

```
## # A tibble: 30 x 3
##    BORO          VIC_AGE_GROUP count
##    <fct>         <fct>         <int>
##  1 BRONX         <18            1017
##  2 BRONX         18-24          3284
##  3 BRONX         25-44          3871
##  4 BRONX         45-64           578
##  5 BRONX         65+              67
##  6 BRONX         UNKNOWN          17
##  7 BROOKLYN      <18            1187
##  8 BROOKLYN      18-24          4161
##  9 BROOKLYN      25-44          5341
## 10 BROOKLYN      45-64           874
## 11 BROOKLYN      65+              93
## 12 BROOKLYN      UNKNOWN          29
```

```
## 13 MANHATTAN     <18              435
## 14 MANHATTAN     18-24           1334
## 15 MANHATTAN     25-44           1874
## 16 MANHATTAN     45-64            294
## 17 MANHATTAN     65+               32
## 18 MANHATTAN     UNKNOWN            7
## 19 QUEENS        <18              354
## 20 QUEENS        18-24           1601
## 21 QUEENS        25-44           2121
## 22 QUEENS        45-64            302
## 23 QUEENS        65+               34
## 24 QUEENS        UNKNOWN           14
## 25 STATEN ISLAND <18               88
## 26 STATEN ISLAND 18-24            297
## 27 STATEN ISLAND 25-44            356
## 28 STATEN ISLAND 45-64             70
## 29 STATEN ISLAND 65+               10
## 30 STATEN ISLAND UNKNOWN            1
```

We can add proportions to get a better sense of how the boroughs compare to each other in age distribution.

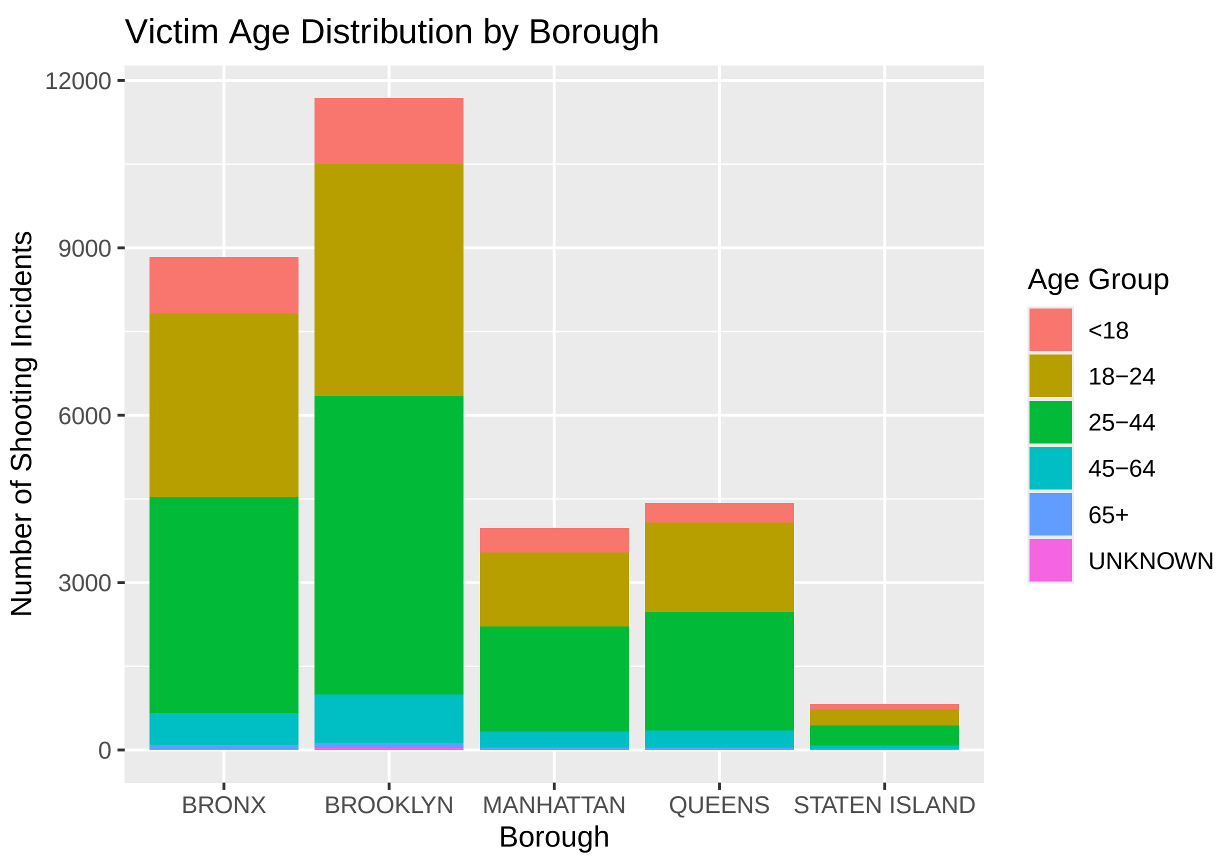
```r
age_boro_pct <- age_boro %>%
  group_by(BORO) %>%
  mutate(proportion = count / sum(count))

print(age_boro_pct, n = Inf)
```

```
## # A tibble: 30 x 4
## # Groups:   BORO [5]
##    BORO          VIC_AGE_GROUP count proportion
##    <fct>         <fct>         <int>      <dbl>
##  1 BRONX         <18            1017    0.115
##  2 BRONX         18-24          3284    0.372
##  3 BRONX         25-44          3871    0.438
##  4 BRONX         45-64           578    0.0654
##  5 BRONX         65+              67    0.00758
##  6 BRONX         UNKNOWN          17    0.00192
##  7 BROOKLYN      <18            1187    0.102
##  8 BROOKLYN      18-24          4161    0.356
##  9 BROOKLYN      25-44          5341    0.457
## 10 BROOKLYN      45-64           874    0.0748
## 11 BROOKLYN      65+              93    0.00796
## 12 BROOKLYN      UNKNOWN          29    0.00248
## 13 MANHATTAN     <18             435    0.109
## 14 MANHATTAN     18-24          1334    0.336
## 15 MANHATTAN     25-44          1874    0.471
## 16 MANHATTAN     45-64           294    0.0739
## 17 MANHATTAN     65+              32    0.00805
## 18 MANHATTAN     UNKNOWN           7    0.00176
## 19 QUEENS        <18             354    0.0800
## 20 QUEENS        18-24          1601    0.362
## 21 QUEENS        25-44          2121    0.479
## 22 QUEENS        45-64           302    0.0682
```

```
## 23 QUEENS        65+              34    0.00768
## 24 QUEENS        UNKNOWN          14    0.00316
## 25 STATEN ISLAND <18              88    0.107
## 26 STATEN ISLAND 18-24           297    0.361
## 27 STATEN ISLAND 25-44           356    0.433
## 28 STATEN ISLAND 45-64            70    0.0852
## 29 STATEN ISLAND 65+              10    0.0122
## 30 STATEN ISLAND UNKNOWN           1    0.00122
```

A visualization of the counts by borough:

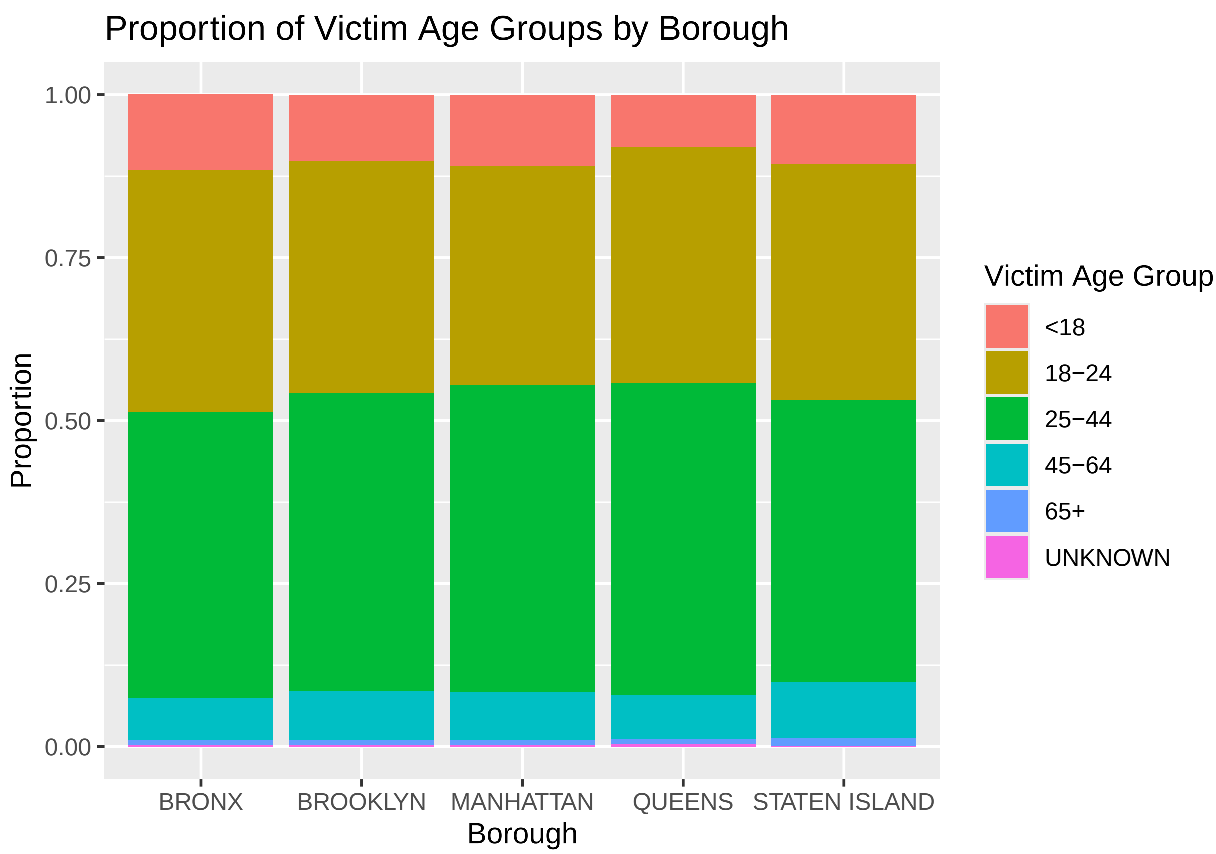
```
age_boro %>%
  ggplot(aes(x = BORO, y = count, fill = VIC_AGE_GROUP)) +
  geom_bar(stat = "identity") +
  labs(title = "Victim Age Distribution by Borough",
       x = "Borough",
       y = "Number of Shooting Incidents",
       fill = "Age Group")
```



A visualization of the proportions of victim age groups in boroughs:

```
age_boro_pct %>%
  ggplot(aes(x = BORO, y = proportion, fill = VIC_AGE_GROUP)) +
  geom_bar(stat = "identity") +
  labs(title = "Proportion of Victim Age Groups by Borough",
       x = "Borough",
```

```r
       y = "Proportion",
       fill = "Victim Age Group")
```



The visualizations raise additional questions of perhaps other victim traits might increase crime numbers in a borough. For instance, I did not explore race or gender, so investigating these traits further might reveal how a victim's profile influences crime. Also, I focused on the victim for this project, but what about the perpetrator of the crime? That might be another consideration, and similarities or differences in perpetrators versus victims might drive different crime volume across boroughs.

Model

I used a Poisson regression model to look at expected shooting crime count. I used borough and victim age group as predictors of crime count.

```r
poisson_model <- glm(
  count ~ BORO + VIC_AGE_GROUP,
  family = poisson,
  data = age_boro)

summary(poisson_model)
```

```
##
## Call:
## glm(formula = count ~ BORO + VIC_AGE_GROUP, family = poisson,
```

```
##     data = age_boro)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)             6.81902    0.02010  339.20   <2e-16 ***
## BOROBROOKLYN            0.27970    0.01410   19.84   <2e-16 ***
## BOROMANHATTAN          -0.79833    0.01910  -41.80   <2e-16 ***
## BOROQUEENS             -0.69111    0.01842  -37.53   <2e-16 ***
## BOROSTATEN ISLAND      -2.37462    0.03647  -65.12   <2e-16 ***
## VIC_AGE_GROUP18-24      1.24284    0.02045   60.77   <2e-16 ***
## VIC_AGE_GROUP25-44      1.48209    0.01996   74.26   <2e-16 ***
## VIC_AGE_GROUP45-64     -0.37478    0.02823  -13.28   <2e-16 ***
## VIC_AGE_GROUP65+       -2.56918    0.06754  -38.04   <2e-16 ***
## VIC_AGE_GROUPUNKNOWN -3.81350    0.12260  -31.11   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 48901.872  on 29  degrees of freedom
## Residual deviance:    77.864  on 20  degrees of freedom
## AIC: 315.88
##
## Number of Fisher Scoring iterations: 4
```

The model chose reference groups BRONX and age group < 18.

Borough

```r
exp(coef(poisson_model)["BOROBROOKLYN"])    # This is doing exp(0.2797)
```

```
## BOROBROOKLYN
##      1.32273
```

```r
exp(coef(poisson_model)["BOROSTATEN ISLAND"])    # This is doing exp(-2.3746)
```

```
## BOROSTATEN ISLAND
##        0.09304958
```

We see BOROBROOKLYN has coefficient = 0.2797. If we exponentiate this, we get about 1.32, which means Brooklyn should be expected to have about 32% more shooting crime incidents than the Bronx, controlling for age group.
BOROSTATEN ISLAND = -2.3746, and if we exponentiate, we get about 0.093 which means Staten Island would have about 0.093 times as many incidents as the Bronx. Said another way, this means Staten Island would have about 90.7% (i.e., 1-0.093 = 0.907) fewer expected count of crimes than the reference group (the Bronx).

Age

For simplicity I just looked at Age 25-44 (the plots showed victims were mostly this group). We exponentiate this group's coefficient.

```r
exp(coef(poisson_model)["VIC_AGE_GROUP25-44"])   # This is doing exp(1.4821)
```

```
## VIC_AGE_GROUP25-44
##           4.402142
```

If we keep borough constant, then we see victims in age 25-44 are expected be seen about 4.4 times as much as the under 18 group (reference group).

Model: Final Thoughts

The null deviance decreased drastically from a big number to a much smaller number (48900 to about 78). This means that borough and victim age group explain a lot of the variation in the shooting crime counts across borough+age combinations.

Project Step 4: Conclusion & Bias

A bias in the data might be that the data represents only reported or documented crime. That in itself might skew the nature of details in the dataset from reality. In essence, we are only getting part of the picture. Some of my personal biases that might have affected my analysis would be that I am from a big and densely populated city, and I personally know several people in my own age group who were victims of crime (though not necessarily shooting-related). So I might want to conclude that people in my age group are more prone to be victims and that dense, big cities should have more crime and more shootings. To mitigate at least my personal biases, I used proportions to calculate the age structure across boroughs, so that even if I am counting volume of incidents and by age group in a particular borough, I can also look at the proportion to see how boroughs compare on the same scale.

Based on the plots, it can be seen that age distribution of victims looks similar across boroughs, which implies that the most common shooting victim age is age group 25-44, wherever the person might be located in New York City. However, there is clearly a difference in volume. I focused on the most and least volume boroughs for sake of comparison (e.g., Brooklyn has the most volume and Staten Island the least). The Poisson model showed that Brooklyn has a positive coefficient (higher expected counts) and Staten Island has a negative one (lower expected counts). This means that if we hold victim age constant across boroughs, then the difference in expected shooting incident counts is due to "something" about the borough. In other words, it means the factor of "borough" influences the expected shooting count rather than any different patterns in victim ages in the borough. Similarly, if we hold borough constant then we can see age 25-44 victims will be seen about 4.4 times as much as the reference group. Finally, we saw the null deviance dropped a lot (48900 to about 78), which supports that borough and victim age group are important predictors on crime count.